Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/31/2005
PI Name:	De Luca, Carlo Ph.D.		
Project Title:	Micro-Gravity Induced Changes in the Control of Muscles		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Physiology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	cjd@bu.edu	**Fax:**	FY 617-353-5737
PI Organization Type:	UNIVERSITY	**Phone:**	617-353-9756
Organization Name:	Boston University - NeuroMuscular Research Center		
PI Address 1:	NeuroMuscular Research Center		
PI Address 2:	19 Deerfield Street, 4th Floor		
PI Web Page:	http://nmrc.bu.edu		
City:	Boston	**State:**	MA
Zip Code:	02215	**Congressional District:**	8
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	99-HEDS-03
Start Date:	05/11/2001	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	6
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:	McCollum, Suzanne	**Contact Phone:**	281 483-7307
Contact Email:	suzanne.g.mccollum@nasa.gov		
Flight Program:	Pre/Post Flight		
Flight Assignment:	In flight development phase (first subjects on STS-114)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Roy, Serge Henri (Boston University NeuroMuscular Research Center)		
Grant/Contract No.:	NCC9-127		
Performance Goal No.:			
Performance Goal Text:			
	The understanding of the neuromuscular basis for microgravity-induced degradation in motor function is essential for the educated design of exercise programs to counter the effects of microgravity as well as make it possible to identify and validate a ground-based model for the phenomenon so that possible countermeasures can be studied in depth on Earth before being attempted in space. The goal of this project is to understand the effects of microgravity on the control of muscles by investigating the regulation of the firing activity of motor units that make up a muscle and the coordination of different muscles responsible for the control of a joint. The following specific aims are being investigated: 1) Understand the effects of microgravity on the control of motor units, 2) Compare the effects of microgravity on two muscles that are used to varying extents in space, in order to assess the potential of muscle use to curtail microgravity-induced disturbances in control properties of motor units, 3) Understand the effects of microgravity on the relative activation patterns of a group		

Task Description:	of muscles controlling one joint, and 4) Determine the course of recovery after re-exposure to Earth gravity. Short duration (Shuttle) crewmembers will serve as subjects in two general sets of procedures performed both pre-flight (L-45, -43, and -41) and post-flight [R+0, (+1 or +2), +3, +5, +7, +14]. Intramuscular electromyographic (EMG) signals will be recorded from a hand muscle, the First Dorsal Interosseous, and a knee extensor muscle, the Vastus Medialis. Surface EMG data will be recorded from muscles surrounding the knee joint. All EMG recordings will be taken while subjects trace specified trajectories displayed on a computer screen. The intramuscular EMG data will be decomposed into motor unit firing trains using a sophisticated set of algorithms constructed with artificial intelligence knowledge-based techniques being developed in our laboratory. Particular attention will be paid to detecting modifications in the control of the firing rate and recruitment behavior of motor units. The surface EMG data will be analyzed to evaluate the effects of microgravity on the coordination of muscles controlling a joint. Particular attention will be directed at comparing the results of this study to similar data obtained by us from elderly subjects to verify whether the effects of microgravity on muscle is similar to that of aging. 1) Determine the neural component to the microgravity-induced deterioration in muscle function and strength, at both the micro and macro levels. This will assist in the design of techniques for monitoring rehabilitative interventions that target the specific alterations in muscle control. These interventions may include exercises that need to be performed in space and those that will be carried out upon return to Earth. 2) Provide the knowledge base necessary to identify a ground-based model for the effects of microgravity (for example, the process of aging). The availability of a validated model for microgravity will facilitate the development of countermeasures by enabling the testing and validation of proposed countermeasures on Earth. 3) By comparing the effects of microgravity on muscles that are used to varying degrees in space, we will evaluate if low-level activity is useful in counteracting some of the effects of microgravity. This knowledge will be important in the design of countermeasures. 4) It is expected that the tracking of the recovery process in this study will identify the distinct effects of the two main the causes for microgravity- related losses in force: atrophy-based and neurally-mediated losses, since muscle force production is expected to recover from these following different timecourses. The knowledge regarding the significance of the contribution of the two components of loss will guide the design of countermeasures since different interventions may be developed to counter the effects of atrophy versus deterioration in neural control. This study aims to answer the basic scientific question regarding the effects of microgravity on the micro and macro control of muscle. By micro control we mean the control of motor units within a muscle; by macro control we mean the control of simultaneously acting muscles. With this information, this study has the potential of providing new strategies for the design of countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The understanding of the neurological modifications that occur in microgravity should provide an accelerated time-scale model for the neurological effect on motor control due to aging. Additionally, the equipment that will be developed for studying the behavior of motor units can be used by neurologists in the clinical environment to assess the severity of diseases such as stroke, Parkinson's disease, etc., as well as the success of various interventions.
Task Progress:	The following progress has been made since the last report period: 1. We have completed Baseline Data Collection (BDC) sessions on an astronaut scheduled to fly on STS-114. Preparations have been made for post-flight testing of this crew member at Kennedy, JSC, and (if needed) Dryden. 2. Updates to the Decomposition Software Package were implemented and integrated into the framework of the existing data acquisition hardware and software. This was done to enhance a "stand alone" portable Decomposition System that will allow us to check the data and determine its decomposition quality within the assigned test period. Knowledge of the quality of the data will enable us to decide if a re-test is necessary, thus reduce the risk of collecting data of low quality. 3. A second hand-restraint apparatus was constructed in-house to be used for collecting data from the first dorsal interosseus muscle of the hand . The newly-designed device overcomes limitations of the originally-proposed hand device. The hand-restraint device accommodates both left and right hand use and provides proper positioning. These improvements are important for providing repeatability and consistency in the measurements of the force and the EMG signals. 4. A third test chair was commissioned for supporting the possibility of post-flight testing on landing day at Dryden. 5. We have continued our progress in developing algorithms for decomposing the surface EMG signal into individual motor unit trains. 6. Evaluative procedures within our laboratory continue to establish the efficacy of the current experimental protocols and equipment as well as provide the opportunity for modifications and practice to improve safety and likelihood of success. The testing involves volunteer subjects that span the anthropometric design specifications for the space shuttle crew. The experimental procedures to be tested are complex and include insertions of needle and fine-wire electrodes, the placement of multiple surface electrodes, recordings during at least 18 different muscle contractions, and multiple practice sessions for operating the re-designed equipment. 7. Basic electrical and mechanical safety tests are being conducted in-house for all instrumentation assigned to this project.
Bibliography Type:	Description: (Last Updated: 12/18/2006)
Abstracts for Journals and Proceedings	Nawab, S.H., Wotiz, R., De Luca, C.J. "Improved Resolution of Pulse Superpositions in a Knowledge-Based System for EMG Decomposition" 26’th International Conference of the IEEE Engineering in Medicine and Biology Society, San Francisco, CA pp 972-982. , Sep-2004
Abstracts for Journals and Proceedings	De Luca C.J., Nawab SH, Adam A, Roark RM, Manto MU, Wotiz R, Srivastava R, Gilmore LD "The Second-Generation Precision Decomposition System For EMG Signals: An Overview" ISEK Congress, Boston pp 180 , Jun-2004
Abstracts for Journals and Proceedings	Adam A and De Luca CJ "Motor Unit Control In Vastus Lateralis Muscle During Fatigue" ISEK Congress, Boston pp. 156 , Jun-2004

Abstracts for Journals and Proceedings	Manto MU, Sauvage C, Adam A, Roark RM, Nawab SH, De Luca, CJ "Motor Unit Firing Patterns In Patients With Cerebellar Ataxia" ISEK Congress, Boston pp. 189 , Jun-2004
Abstracts for Journals and Proceedings	Roark RM, Adam A, Zalvan CH, Manto MU, Nawab SH, De Luca CJ "Clinical Evaluation Of Laryngeal Muscle Dysfunction Using Advanced EMG Decomposition" ISEK Congress, Boston pp. 158 , Jun-2004
Abstracts for Journals and Proceedings	Wotiz, R., De Luca, C.J., Nawab, S.H "Second Generation Precision Decomposition: Solutions" ISEK Congress, Boston pp. 159 , Jun-2004
Abstracts for Journals and Proceedings	Nawab, S.H., Wotiz, R., De Luca, C.J. "Second Generation Precision Decomposition: Challenges" ISEK Congress, Boston pp. 160 , Jun-2004
Abstracts for Journals and Proceedings	Manto M.U., Sauvage C., Adam A., Roark R.M., Nawab S.H., De Luca CJ. "Motor Unit Firing Patterns in Patients with Cerebellar Ataxia" Society for Neuroscience, Washington, D.C. Abstract Viewer/Itinerary Planner Program No. 189.5. , Oct-2004
Abstracts for Journals and Proceedings	Roark R.M., Adam A., Zalvan C.H., Manto M.U., Nawab S.H., De Luca CJ "Multiple Motor Unit Recordings of Intra-Laryngeal Muscles" Society for Neuroscience, Washington, D.C. Abstract Viewer/Itinerary Planner, Program No. 189.17 , Oct-2004
Abstracts for Journals and Proceedings	De Luca, C.J., Roy, S.H., Adam, A., Gilmore, L.D "Microgravity Induced Changes in the Control of Motor Units" NASA Bioastronautics Investigator's Workshop, Galveston, TX Abstract 96, pp. 24 of Program , Jan-2005